

Introduction to distribution modelling

QGIS R Workshop NHM 2025

Olav Skarpaas

Geo-ecology research group (GEco)

Distribution modelling – what and why?

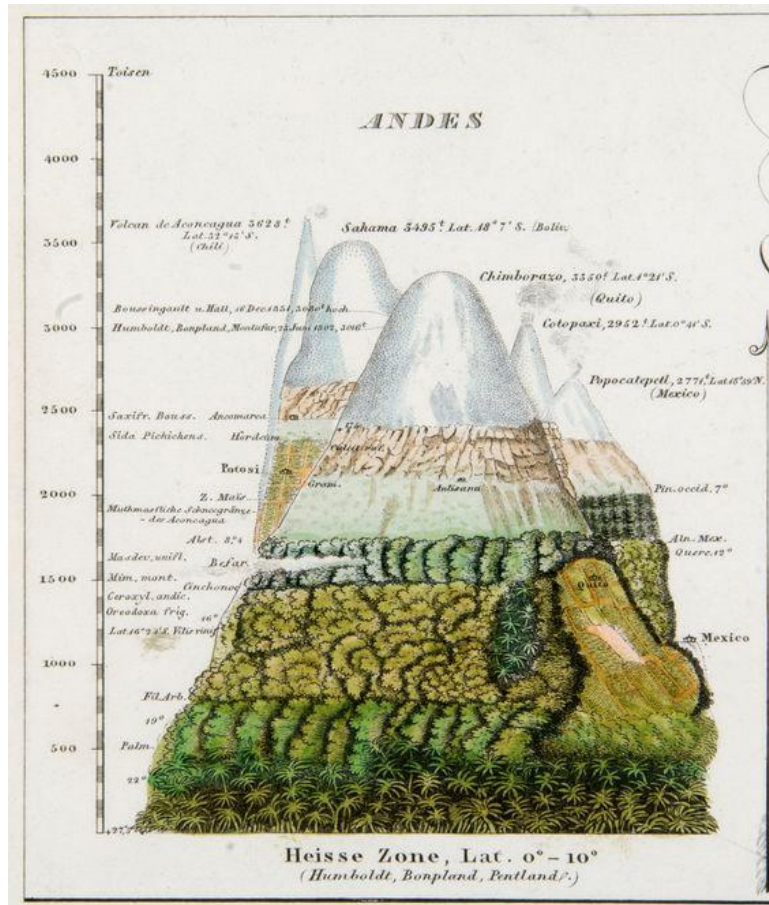
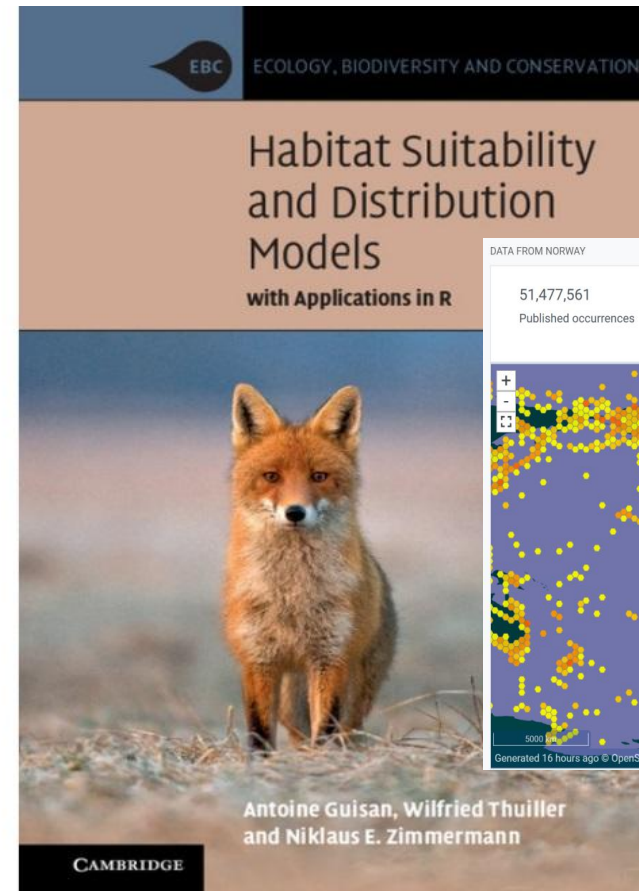


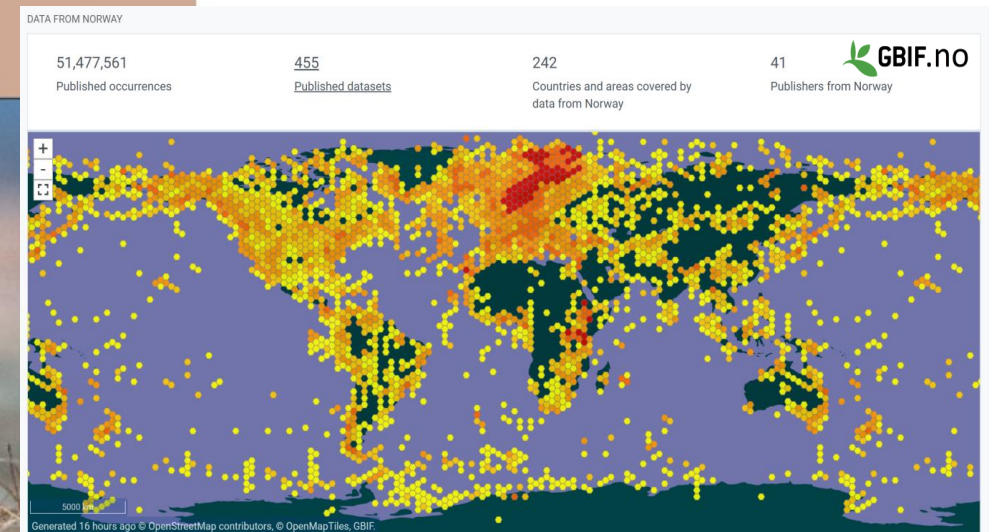
Figure from Humboldt's Berghaus Atlas (1845-62)

photo by David Bressan

<https://www.forbes.com/sites/davidbressan/2016/09/29/alexander-von-humboldts-contributions-to-geology/#78a43e2e776f>



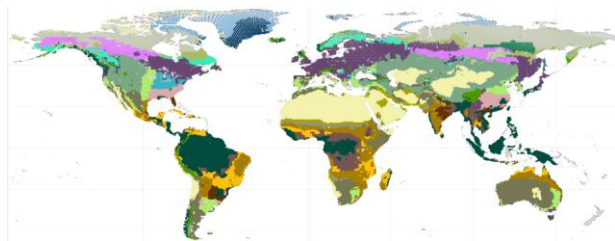
Methods: e.g. Guisan et al. 2017



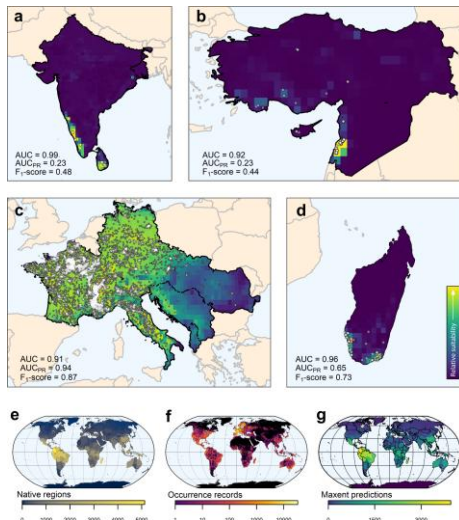
Abundant data: e.g. GBIF

Distribution modelling – what and why?

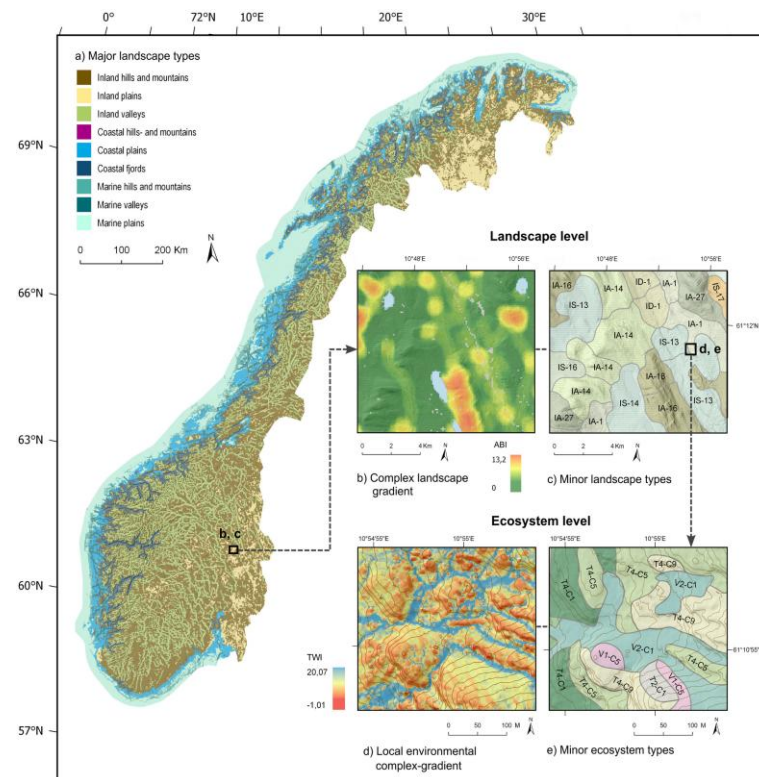
Examples of distribution modelling by GEco and allies



Global map of vegetation types (Beigaite et al. 2022)



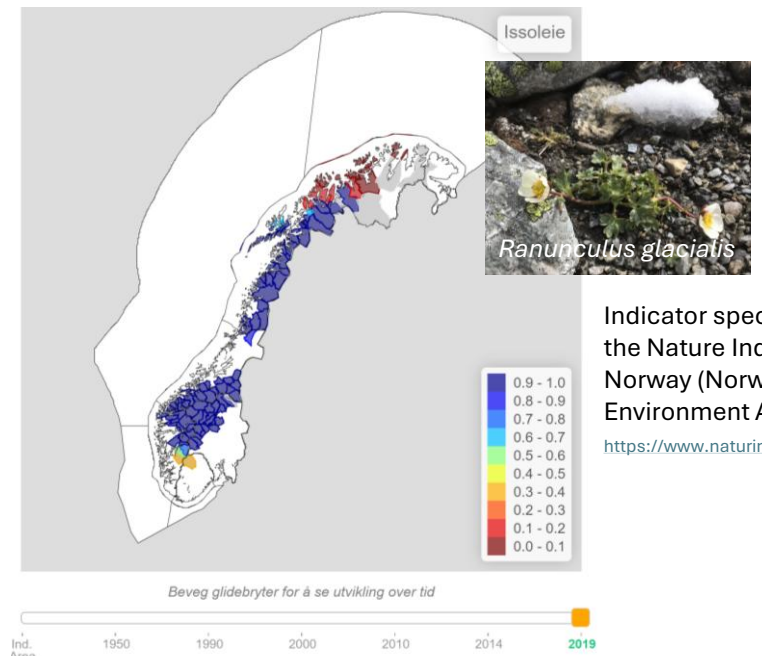
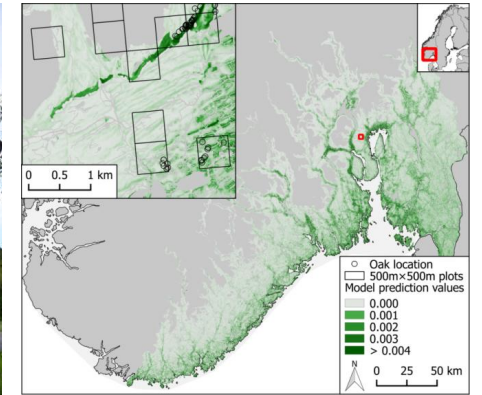
Range estimates for >27000 red-listed plants (Borgelt et al. 2022)



National, regional and local maps of landscape and ecosystem types (NiN / EcoSyst; Halvorsen et al. 2020)



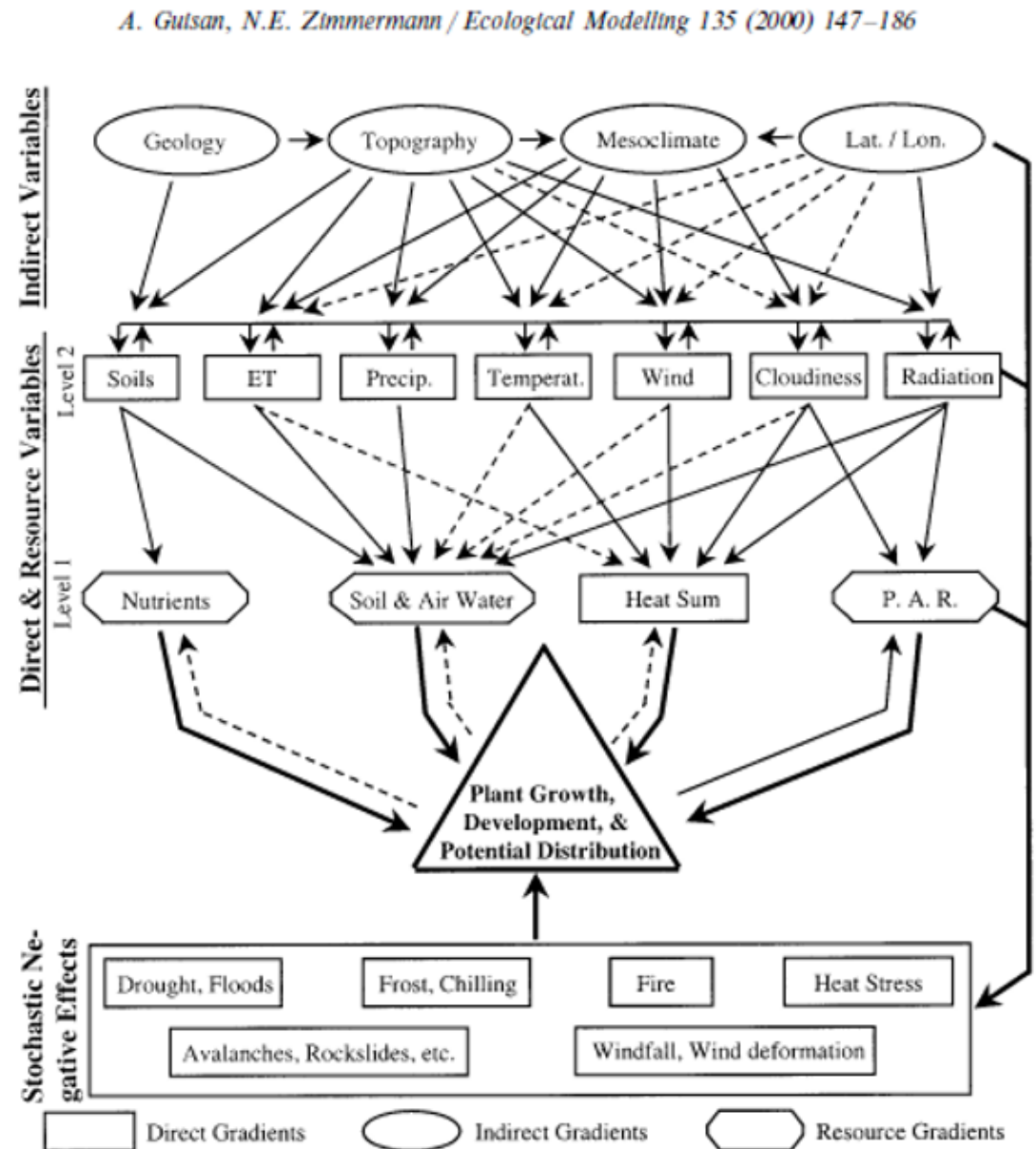
Priority habitats: veteran trees (Skarpaas et al. 2017)



Indicator species in the Nature Index for Norway (Norwegian Environment Agency, <https://www.naturindeks.no/>)

Target, direct and indirect variables

- Many variables interact in complex ways (causal, covariation)
- Direct variables often work better in DMs
- Choice of variables depends on
 - Modelling purpose
 - Available data
 - Modelling method



Modelling process

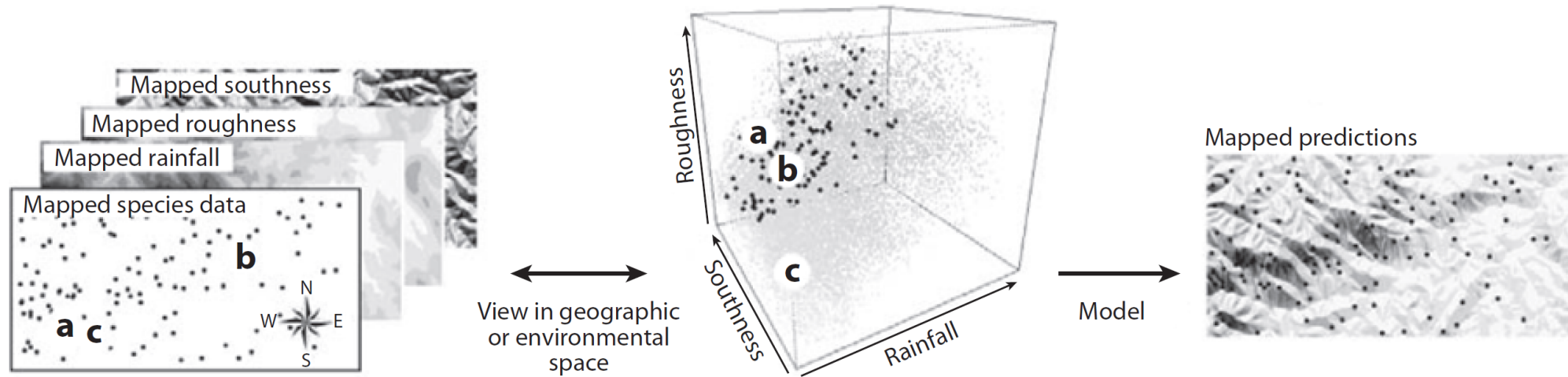


Figure 1

The relationship between mapped species and environmental data (*left*), environmental space (*center*), and mapped predictions from a model only using environmental predictors (*right*). Note that inter-site distances in geographic space might be quite different from those in environmental space—*a* and *c* are close geographically, but not environmentally. The patterning in the predictions reflects the spatial autocorrelation of the environmental predictors.

Modelling process

A. Guisan, N.E. Zimmermann / Ecological Modelling 135 (2000) 147–186

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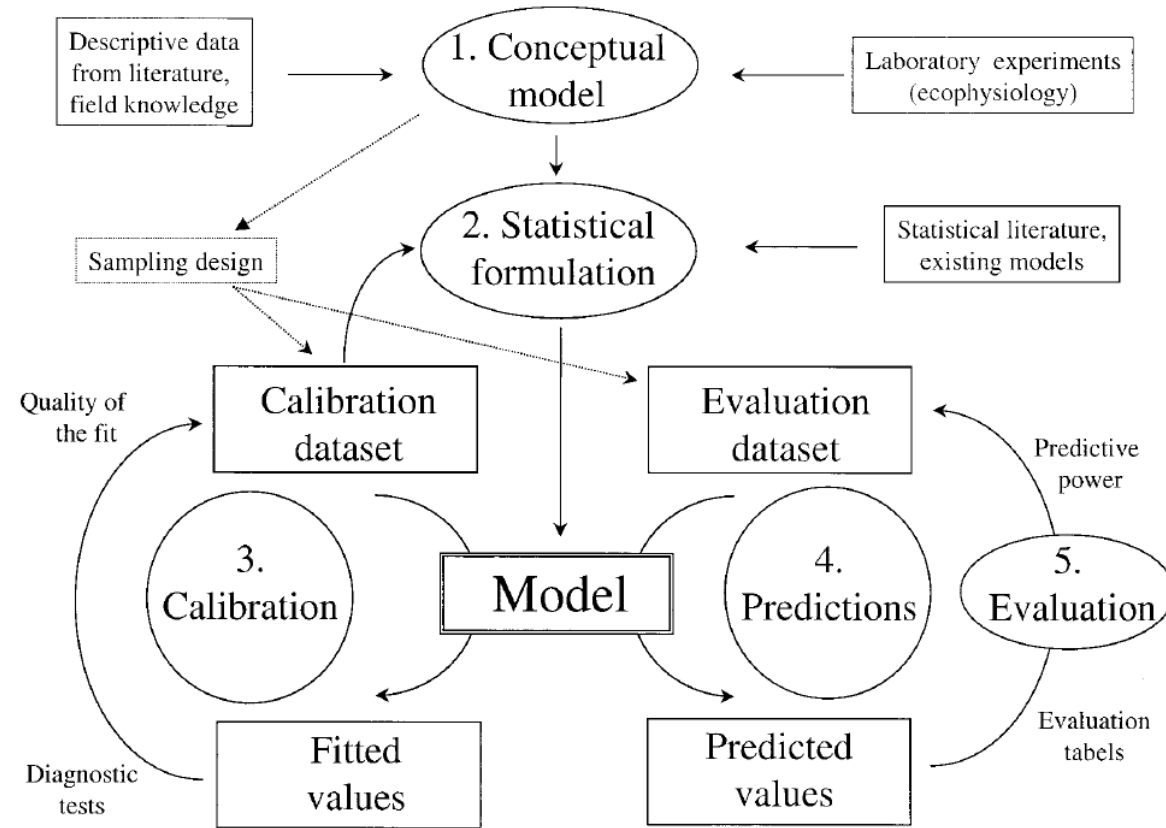


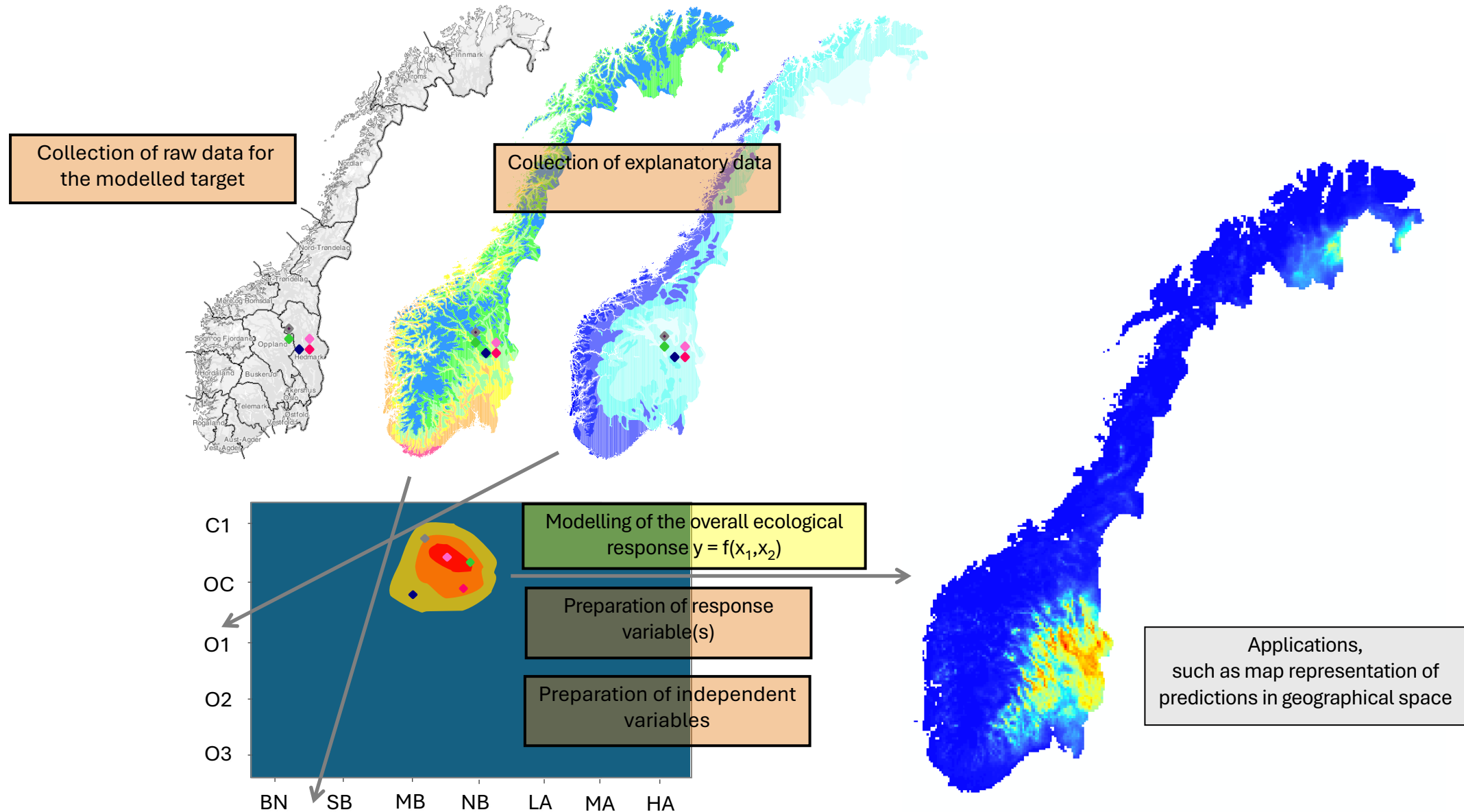
Fig. 1. Overview of the successive steps (1–5) of the model building process, when two data sets – one for fitting and one for evaluating the model – are available. Model evaluation is either made: (a) on the calibration data set using bootstrap, cross-validation or Jack-knife techniques; (b) on the independent data set, by comparing predicted to observed values using preferentially a threshold-independent measure (such as the ROC-plot approach for presence/absence models).

General steps – with some overlap/dependencies

1. Conceptual model formulation and data
2. Formulate statistical model
3. Fit model to training data
4. Validate fitted model
 - Assess diagnostics based on **training data**
5. Evaluate fitted model
 - Assess diagnostics based on **evaluation data**

Main focus today

Distribution modelling in practice



Modelling approaches

- Many approaches available, varying in
 - data handling (presence only, presence-background, presence-absence)
 - statistical methodology
 - software availability
 - performance

(see e.g. Guisan & Zimmermann 2000, Elith et al. 2006, Pearson 2010)

Method(s) ¹	Model/software name ²	Species data type	Key reference/URL
Gower Metric	DOMAIN*	presence-only	Carpenter et al., 1993 http://www.cifor.cgiar.org/docs/_ref/research_tools/domain/ http://diva-gis.org
Ecological Niche Factor Analysis (ENFA)	BIOMAPPER*	presence and background	Hirzel et al., 2002 http://www2.unil.ch/biomapper/
Maximum Entropy	MAXENT*	presence and background	Phillips et al., 2006 http://www.cs.princeton.edu/~schapire/maxent/
Genetic algorithm (GA)	GARP ³ *	pseudo-absence ⁴	Stockwell and Peters, 1999 http://www.lifemapper.org/desktopgarp/
Artificial Neural Network (ANN)	SPECIES	presence and absence (or pseudo-absence)	Pearson et al., 2002
Regression: generalized linear model (GLM), generalized additive model (GAM), boosted regression trees (BRT), multivariate adaptive regression splines (MARS)	Implemented in R ⁵	presence and absence (or pseudo-absence)	Lehman et al., 2002 Elith et al., 2006 Leathwick et al., 2006 Elith et al., 2007
Multiple methods	BIOMOD	presence and absence (or pseudo-absence)	Thuiller, 2003
Multiple methods	OpenModeller	depends on method implemented	http://openmodeller.sourceforge.net/

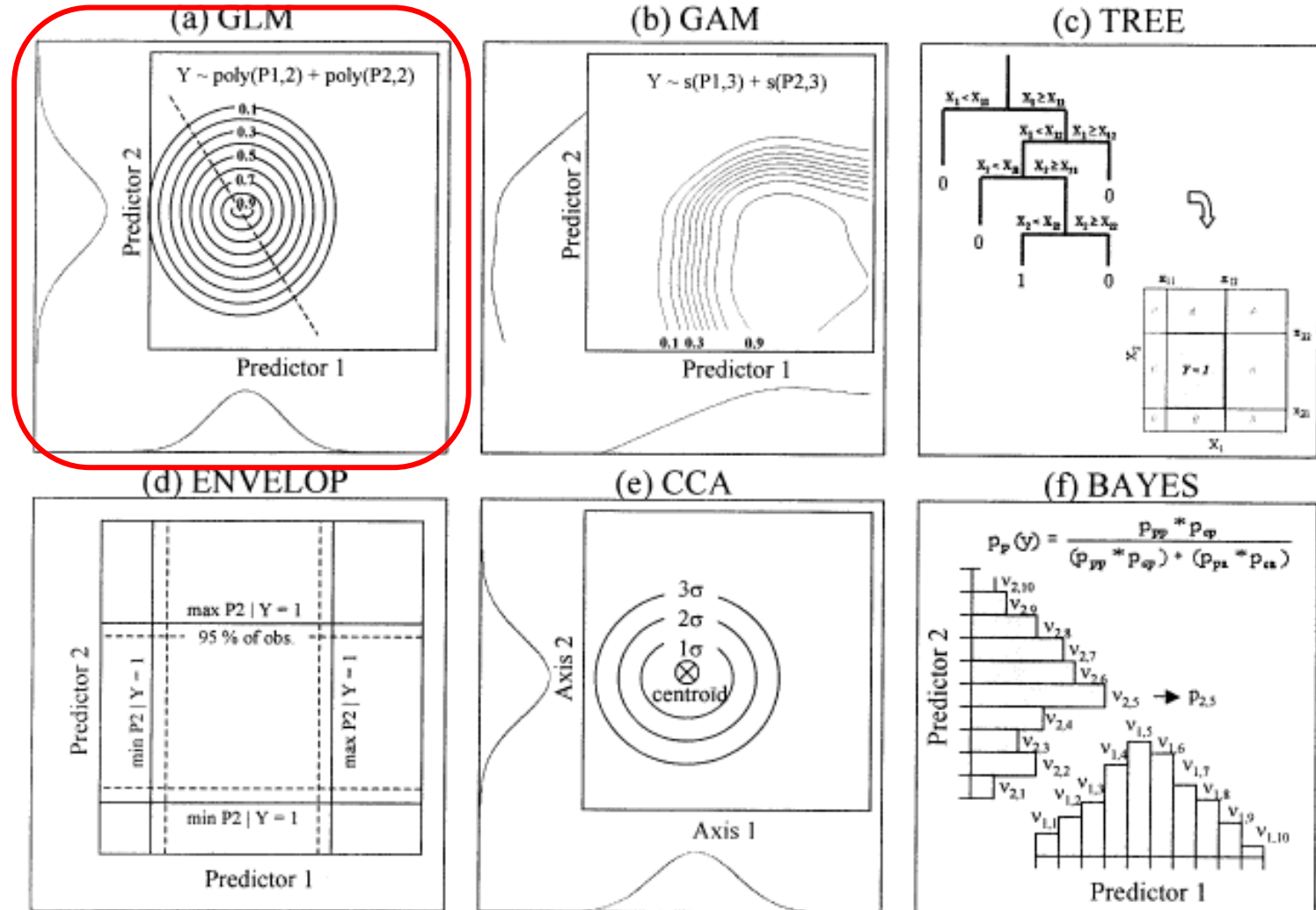
¹ 'Method' refers to a statistical or machine-learning technique. ²'Model/software name' refers to a name (or acronym) given to a published model that implements the method(s) stated. Software to implement the method for species' distribution modeling is readily available at no cost for those models marked with an asterisk (*); other models are available at the discretion of the author(s). ³The genetic algorithm for rule-set prediction (GARP) includes within its processing multiple methods, including GLM. ⁴Note that Pseudo-absence here refers to the sampling approach implemented in the GARP software; in principle, any presence-absence method can be implemented using pseudo absences. ⁵R is a freely available (at no cost) software environment for statistical computing and graphics (<http://www.r-project.org/>). Based in part on Elith et al. (2006) and Guisan and Thuiller (2005).

Modelling approaches

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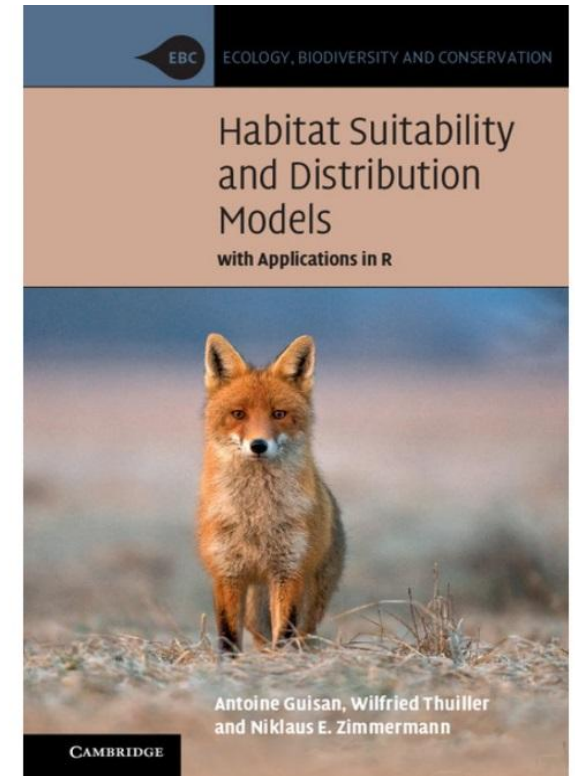
A. Gutsan, N.E. Zimmermann / Ecological Modelling 135 (2000) 147–186

Main focus today



General assumptions (in «classical» regression-based DM)

- Target in equilibrium with environment
- Availability of all important predictors
- Appropriateness of target observations
- Appropriateness of statistical methods
- Predictors measured without error
- Unbiased target data
- Independence of target observations



Guisan et al. 2017

A basic approach for presence-absence data: Logistic regression

- Generalized linear models (GLM)
 - Parametric, flexible regression models by which the response variable is first transformed with a **link function** g , thereafter maximum likelihood estimation is used to model the transformed response as a **linear function** of predictor variables x_i and parameters β_i
 - GLM with logistic link and binomial errors, i.e. **logistic regression**, is relevant for SDM: Probability of presence of the target species (p), modelled with presence/absence data as a function of one to several predictors (x)

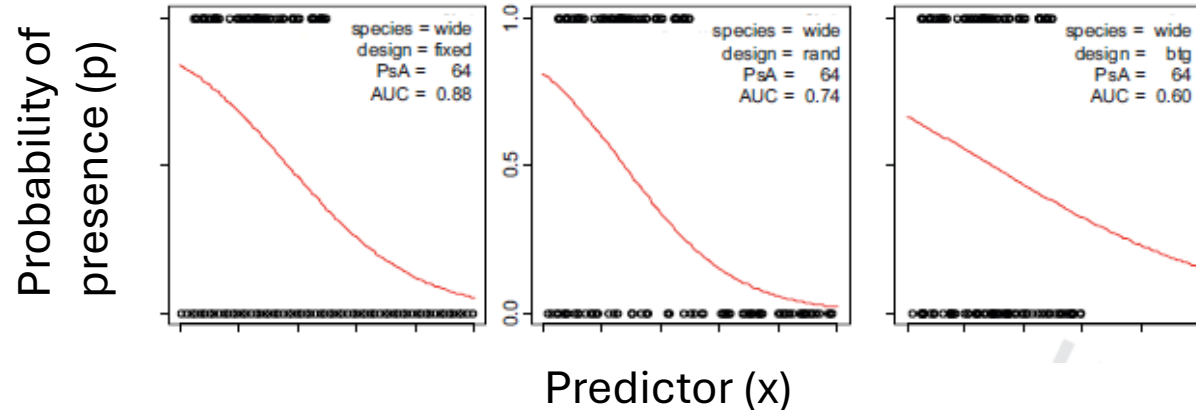
$$g(p) = \ln \frac{p}{1-p} = \beta_0 + \sum_{i=1}^m \beta_i x_i$$

(e.g. McCullagh & Nelder 1989)

A basic approach for presence-absence data: Logistic regression

- Generalised linear models (GLM) with logistic link function and binomial errors: **logistic regression**
 - Parameters (betas) typically estimated with **maximum likelihood**
 - To obtain estimates for probability of presence of the target (p), we have to back-transform the results:

$$\ln \frac{p}{1-p} = \beta_0 + \sum_{i=1}^m \beta_i x_i \Leftrightarrow p = \frac{e^{\beta_0 + \sum_{i=1}^m \beta_i x_i}}{1 + e^{\beta_0 + \sum_{i=1}^m \beta_i x_i}}$$



Adapted from
Stokland et al. 2011

GLM example: Veteran oaks (*Quercus* spp.)

- Biodiversity hotspot: >100 redlisted insects ++



Protætia marmorata
Photo A Sverdrup-Thygeson

Veteran oaks (*Quercus* spp.)

- Occur in forests and open landscapes in Norway
- Protected in open landscapes, but not in productive forest
- Where and how many?



Photos A Sverdrup-Thygeson

Veteran oaks (*Quercus* spp.)

Received: 3 March 2017 | Revised: 13 June 2017 | Accepted: 28 June 2017

DOI: 10.1002/ece3.3305

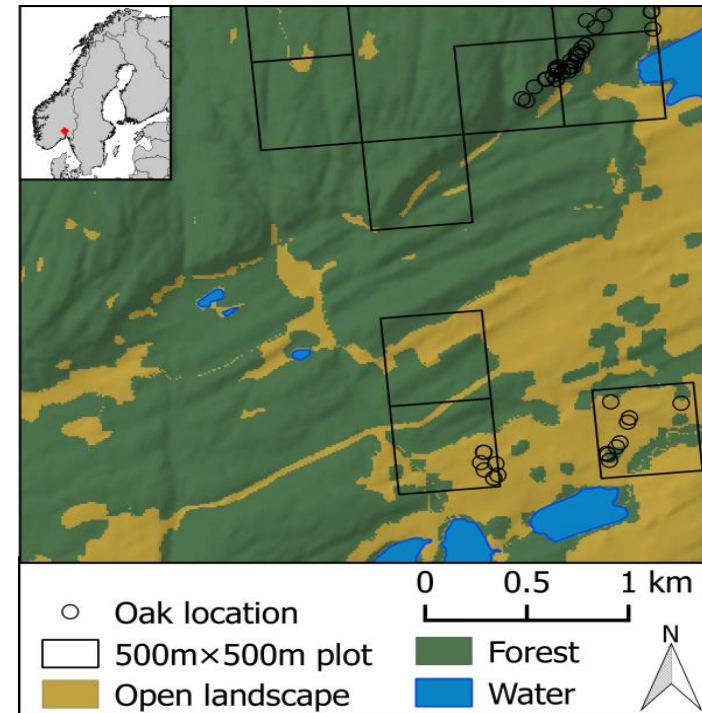
ORIGINAL RESEARCH

WILEY *Ecology and Evolution*

Prediction of biodiversity hotspots in the Anthropocene: The case of veteran oaks

Skarpaas et al. 2017

- Stratified sample of presence-absence data from monitoring pilot
- GLMs
 - Single regression
 - Multiple regression
 - Multiple models



Veteran oaks (*Quercus* spp.)

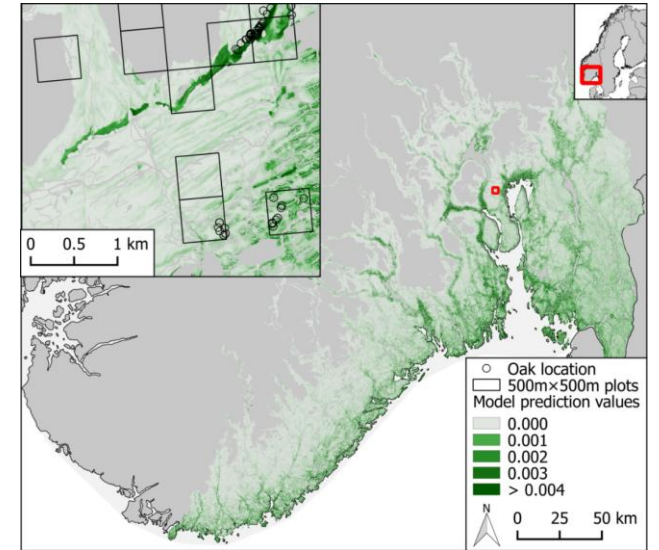
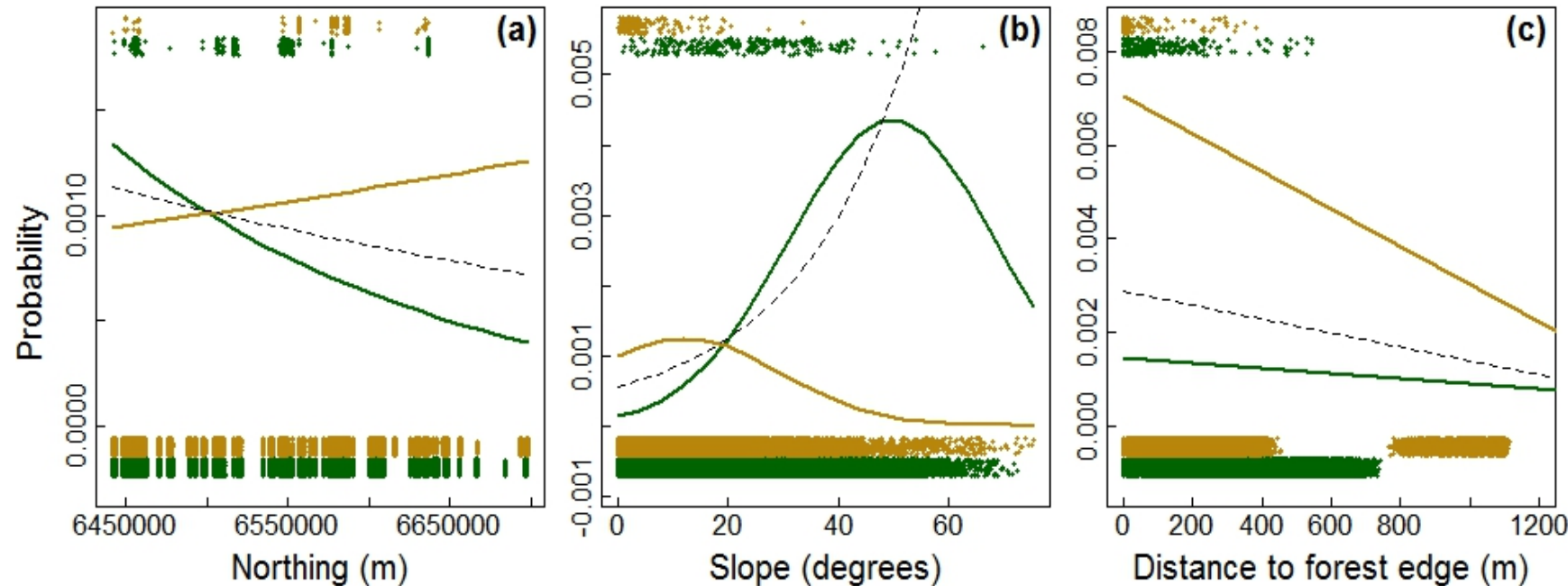
Initial variable screening,
correlation tests, etc.:

80 -> 13 variables

No.	Variable	Definition (units/scale)	Related patterns and processes
1	X	West-east coordinate (m)	Oceanic-continental (moisture) gradient
2	Y	South-north coordinate (m)	Nemoral-boreal (temperature) gradient
3	Z	Elevation above sea level (m)	Correlated with air temperature, marine deposits, land use, etc.
4	S	Slope (degrees)	Correlated with insolation time, ground conditions and forestry activity
5	A	Aspect (northness)	Correlated with radiation sum
6	TWI	Terrain wetness index	Wetness indicator based on terrain and water flow from above
7	K	Fishers K (100 m radius)	Expresses terrain ruggedness, i.e. topographic variability
8	FA	Area of forest (patches >20 m across) within 1 km	Negatively related to intensive agriculture and urban areas
9	FD	Distance to forest edge (m)	Correlated with distance to open landscape
10	RD	Distance to road (m; water as a hard barrier)	Related to ease of access for logging, and for ornamental plantings (e.g. avenues)
11	WD	Distance to water (m)	Related to ease of access for past logging
12	T	Dominant tree class (T31: spruce, T32: pine, T33: deciduous)	Dominant tree species in forests
13	P	Productivity class (site index; P12_13: 12-13 m, etc.)	Represents productivity as reflected in height of the dominant tree species

Veteran oaks (*Quercus* spp.)

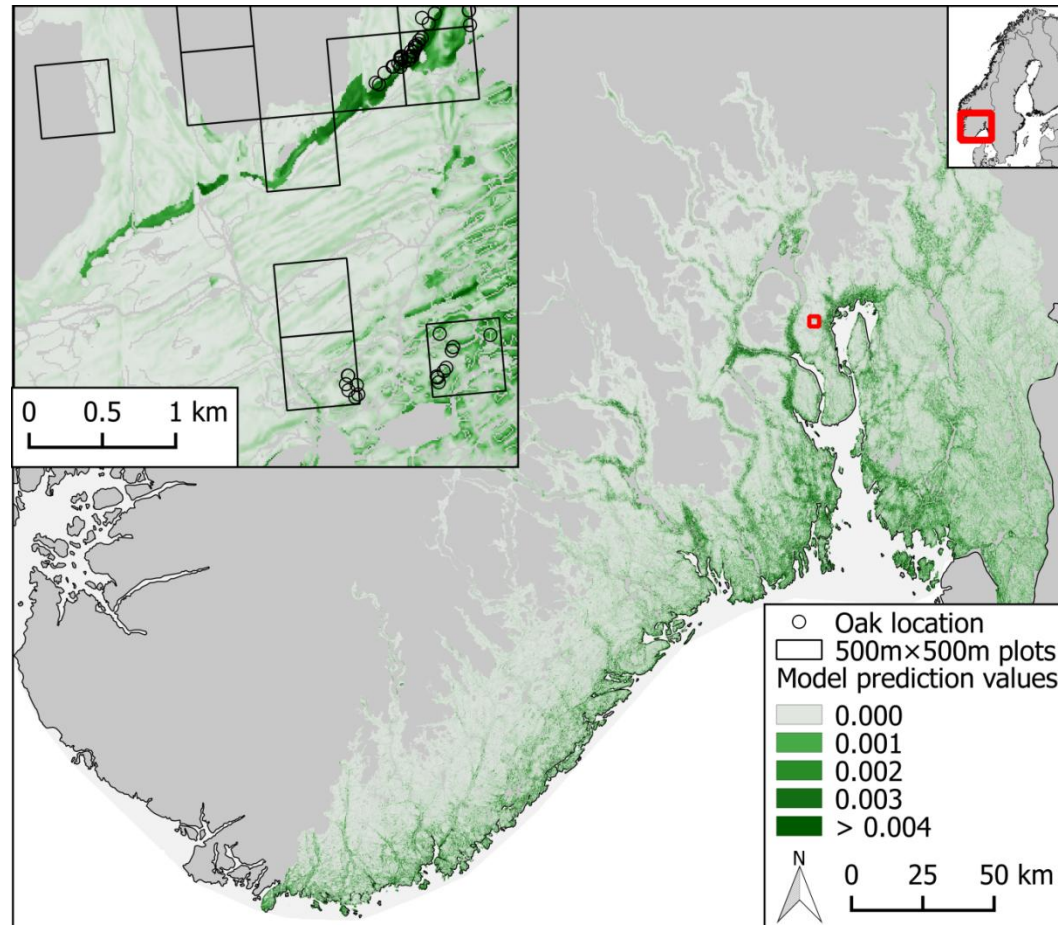
- Different responses in forests (green) and open landscapes (yellow)
- Masked / unrealistic responses when considered together (dashed)



Skarpaas et al. 2017

Veteran oaks (*Quercus* spp.)

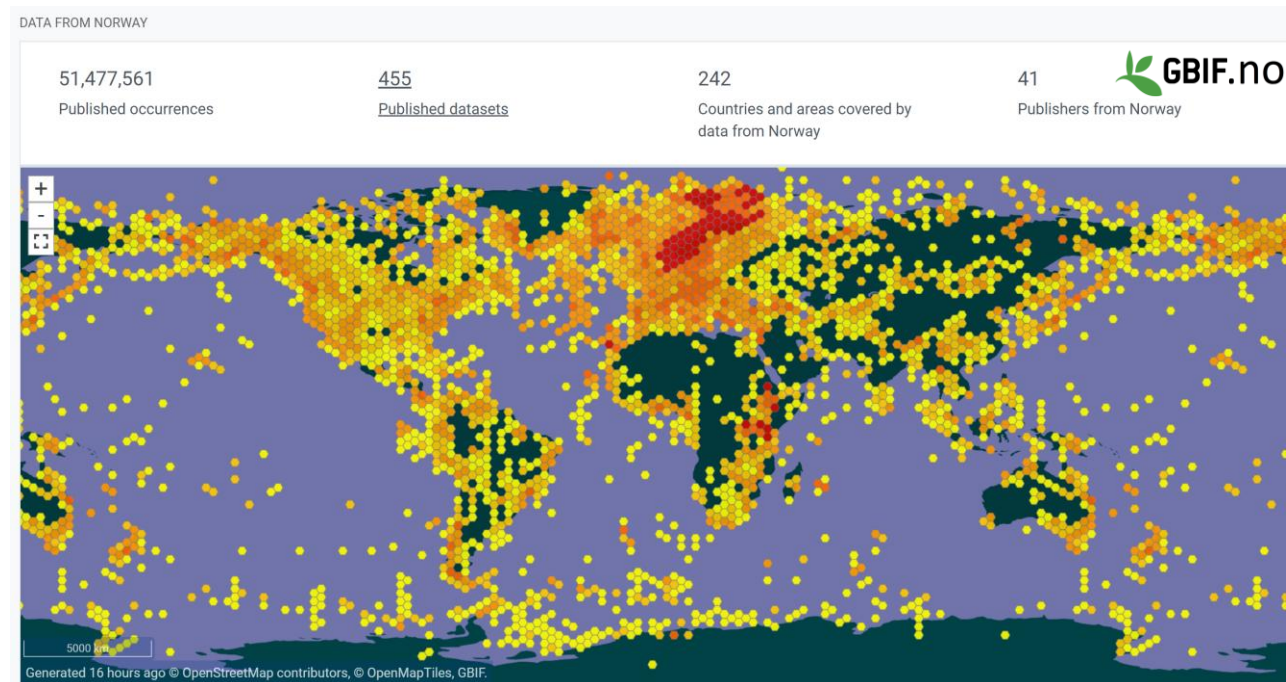
- Forest and open landscape model predictions combined



Skarpaas et al. 2017

OK, so logistic regression works for presence-absence data

- But what if we don't have absence data?



GBIF: Mainly presence data, but some absences as well

There are ways to generate pseudo-absences

- One entry-point to the literature:

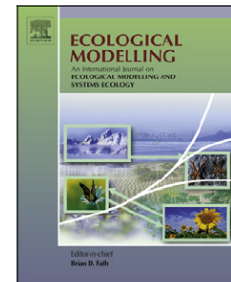
Ecological Modelling 312 (2015) 166–174



Contents lists available at [ScienceDirect](#)

Ecological Modelling

journal homepage: www.elsevier.com/locate/ecolmodel



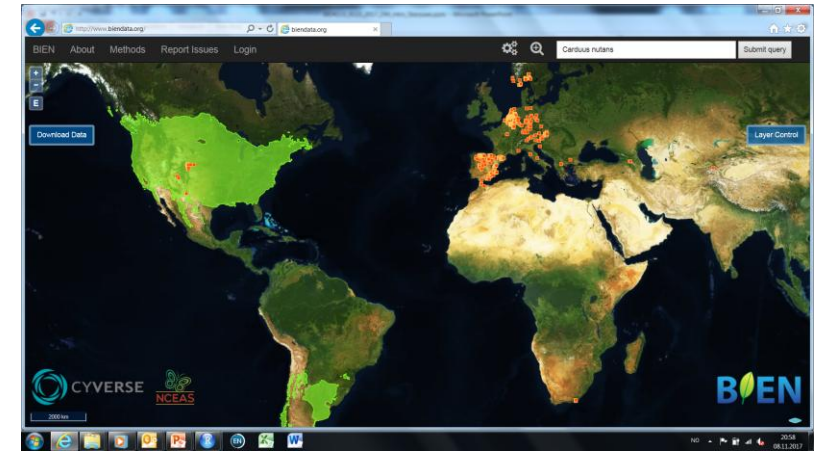
A framework for species distribution modelling with improved pseudo-absence generation



Maialen Iturbide^{a,*,1}, Joaquín Bedia^a, Sixto Herrera^{b,c}, Oscar del Hierro^d, Miriam Pinto^d, Jose Manuel Gutiérrez^a

A popular method for presence data: MaxEnt

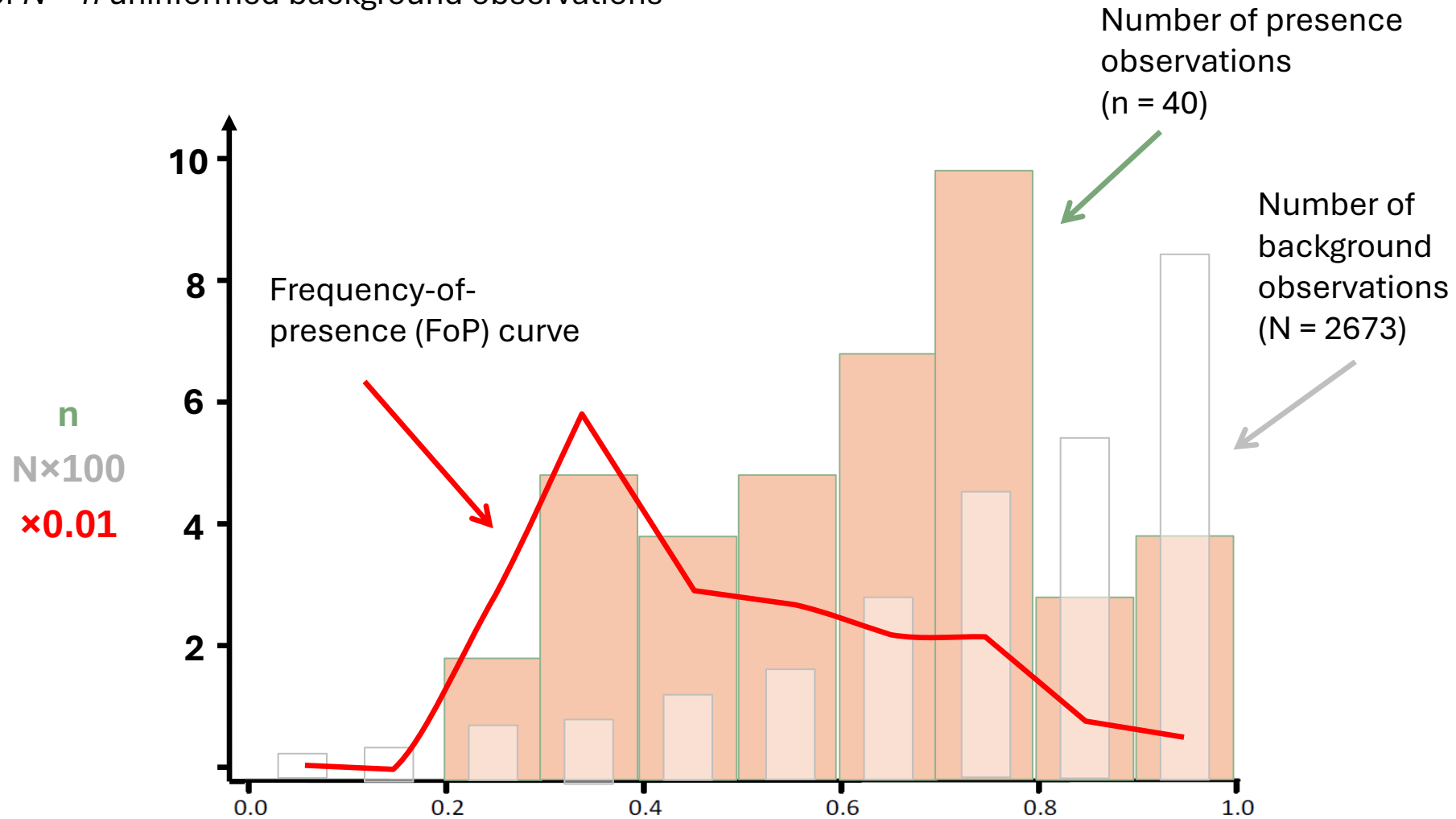
- Regarded as one of the best DM methods (e.g., Elith et al. 2006)
- MaxEnt is a statistical modelling method based upon the '**m**aximum **e**ntropy principle' (Jaynes 1957a, 1957b)
- Core idea: «to estimate a target probability distribution by finding the probability distribution of maximum entropy (i.e. that is most spread out, i.e. closest to uniform), subject to a set of constraints that represent our incomplete information about the target distribution.» (Phillips et al. 2006)
- MaxEnt is often described as a machine learning method (e.g., Phillips et al. 2004, Phillips & Dudík 2008), but can also be described as a **maximum likelihood estimation method** (e.g., Halvorsen et al. 2015)



Thousands of maxent models (e.g. BIEN)

MaxEnt – a simple explanation

- MaxEnt models the frequency of presence (FoP) relative to a set of background observations, typically
 - the set of n observed presences *and*
 - a set of $N - n$ uninformed background observations



MaxEnt – a simple explanation (2D)

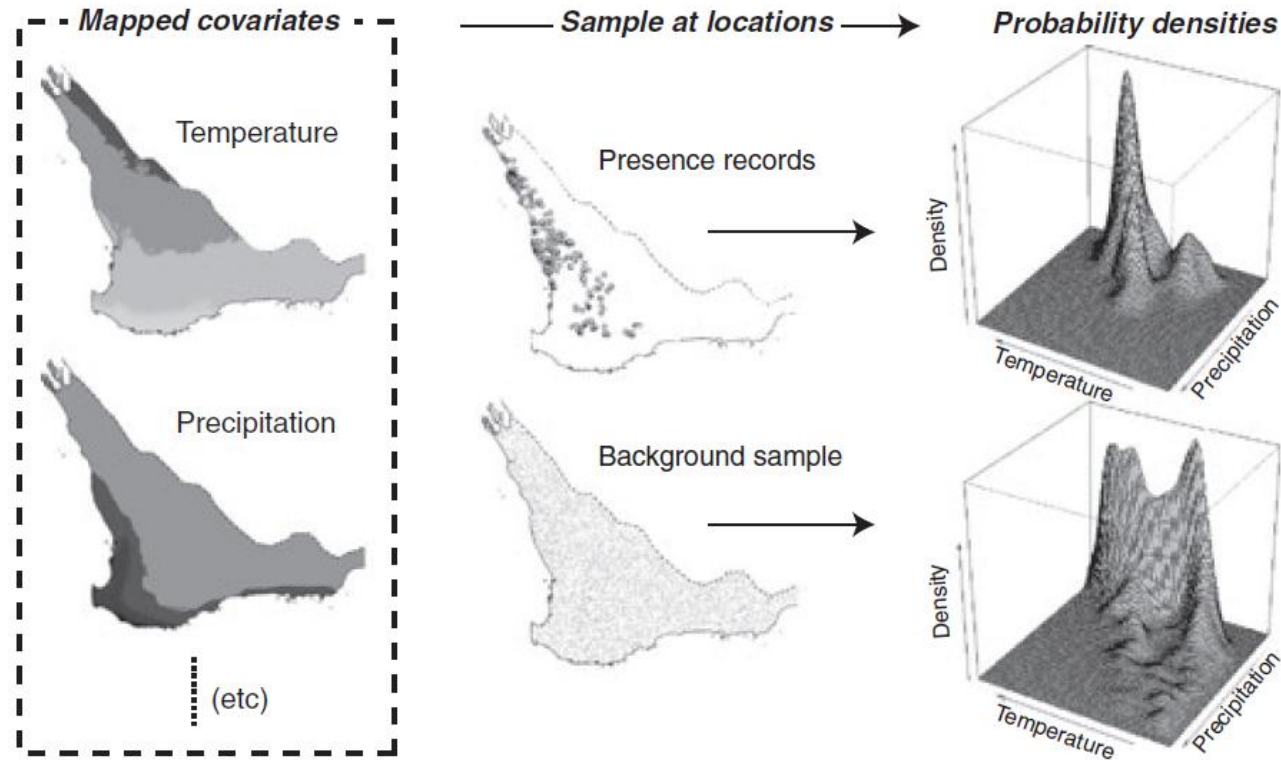
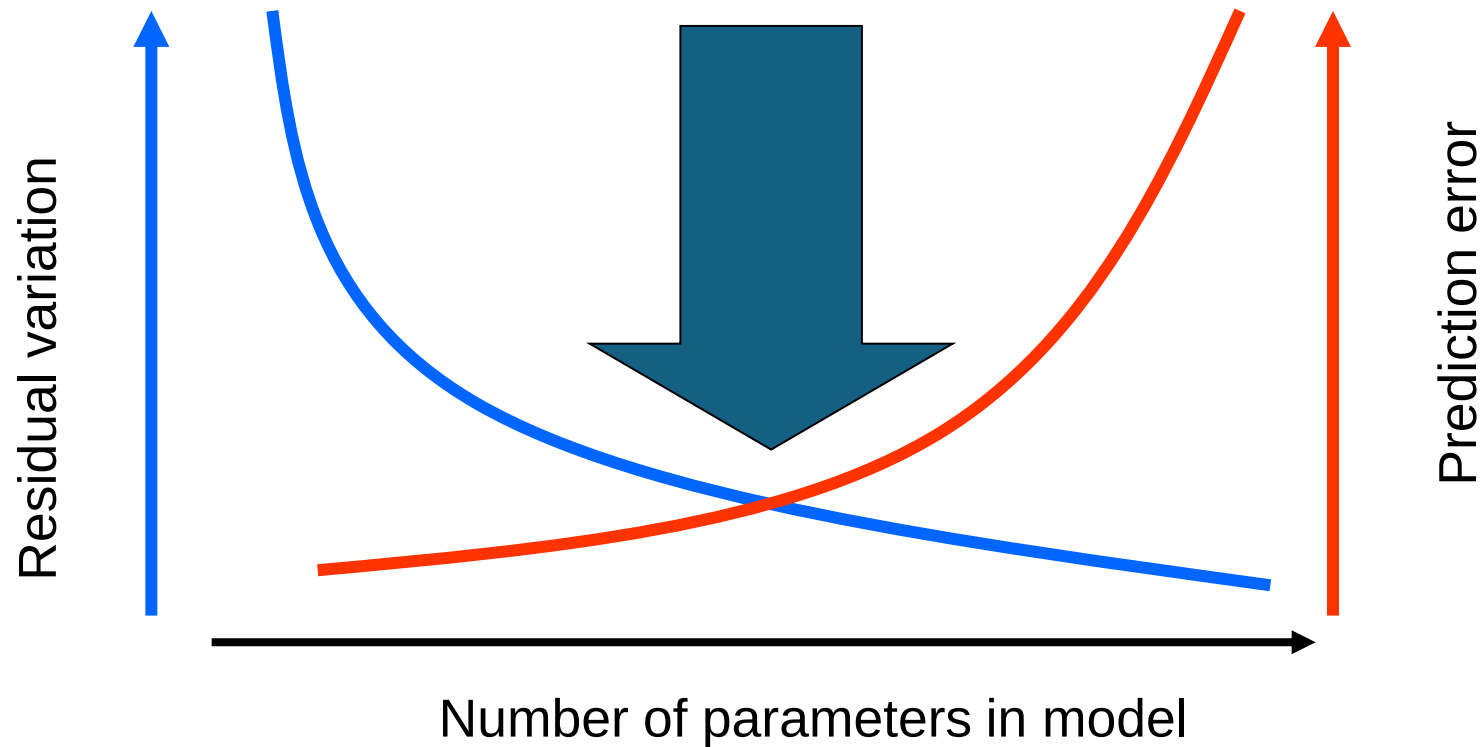


Figure 1 A diagrammatic representation of the probability densities relevant to our statistical explanation, using data presented in case study 1. The maps on the left are two example mapped covariates (temperature and precipitation). In the centre are the locations of the presence and background samples. The density estimates on the right are not in geographic (map) space, but show the distributions of values in covariate space for the presence (top right) and background (bottom right) samples. These could represent the densities $f_1(\mathbf{z})$ and $f(\mathbf{z})$ for a simple model with linear features.

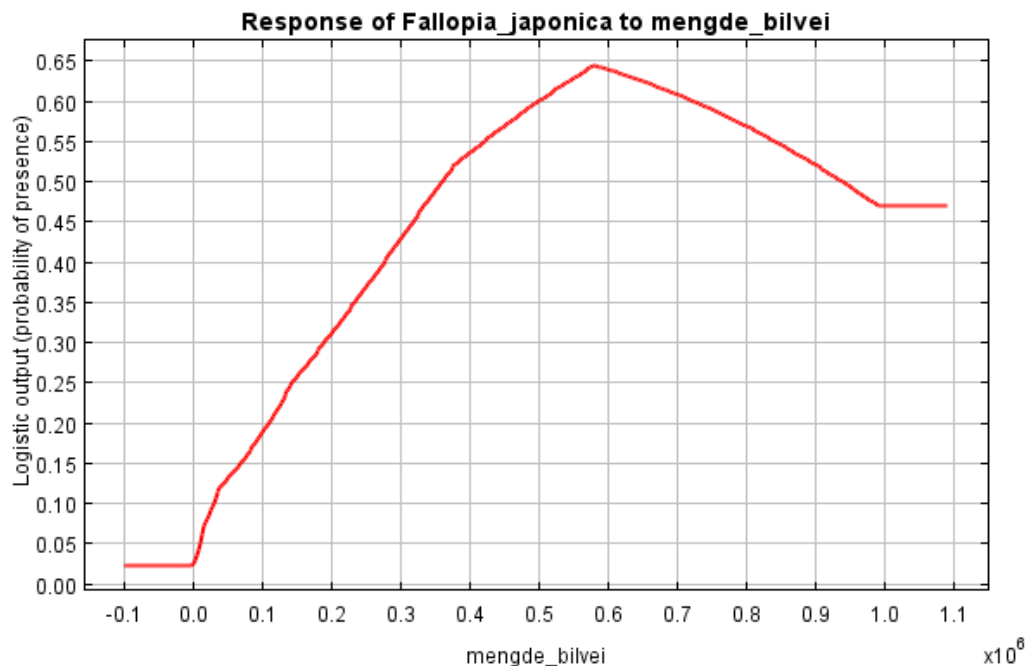
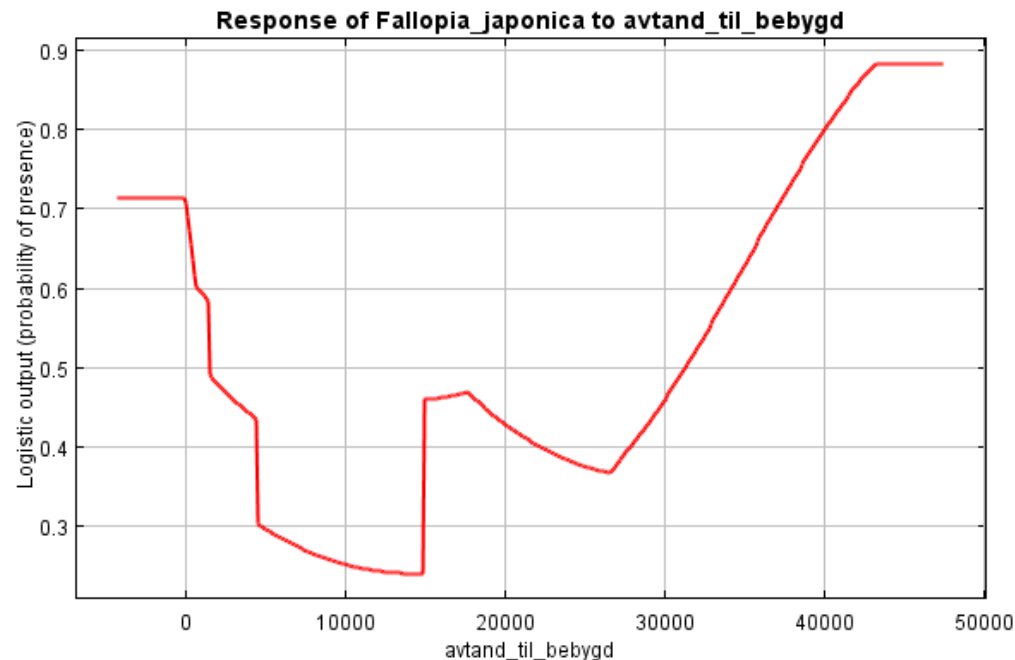
Challenge: overfitting

The principle of **parsimony**:

Trade-off between model complexity and variation explained; the model that best balances these two properties, is the best model



Ockham's Razor: among several plausible explanations for a phenomenon, the simplest is the best



$$q_i = g_{\theta}(X_i) = e^{\theta_0 + \sum_{k=1}^m \theta_k x_{ki}}$$

```
avtand_til_beygd, 0.0, 0.0, 43129.0
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mengde_dyrka, -0.5765333571177134, 0.0, 1499.0
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pca2int, -0.9565040652456436, -0.004100000020116568, 0.0044999998062849045
relieff, 0.0, 0.0, 1363.0
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numBackgroundPoints, 10063
entropy, 6.713432407837414
```

Complex model with 51 parameters
(plus intercept), logistic output

Some R tools for model fitting, selection, validation and evaluation

- R packages
 - `base: functions glm, summary, plot, etc.`
 - `raster, terra: useful for spatial data, including spatial model predictions`
 - Examples of packages with tailor-made tools for distribution modelling:
`biomod2, dismo, sdm, ENMTools, ENMeval, MIAMaxent`

Other methods

- GAM, regression trees, etc.
- Occupancy models
- Mark-recapture models
- Ensemble models
- Hierarchical models
- Joint species distribution models
- Dynamic vegetation models
- Demographic/mechanistic distribution models
- ...

(several topics are covered in the distribution modelling course at UiO, BIOS 5211/9211)

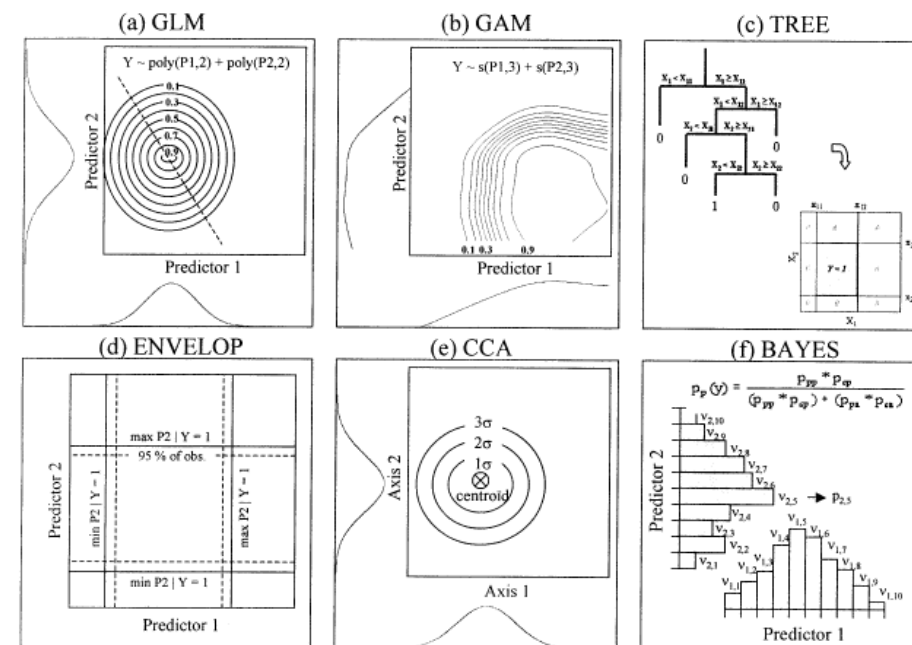
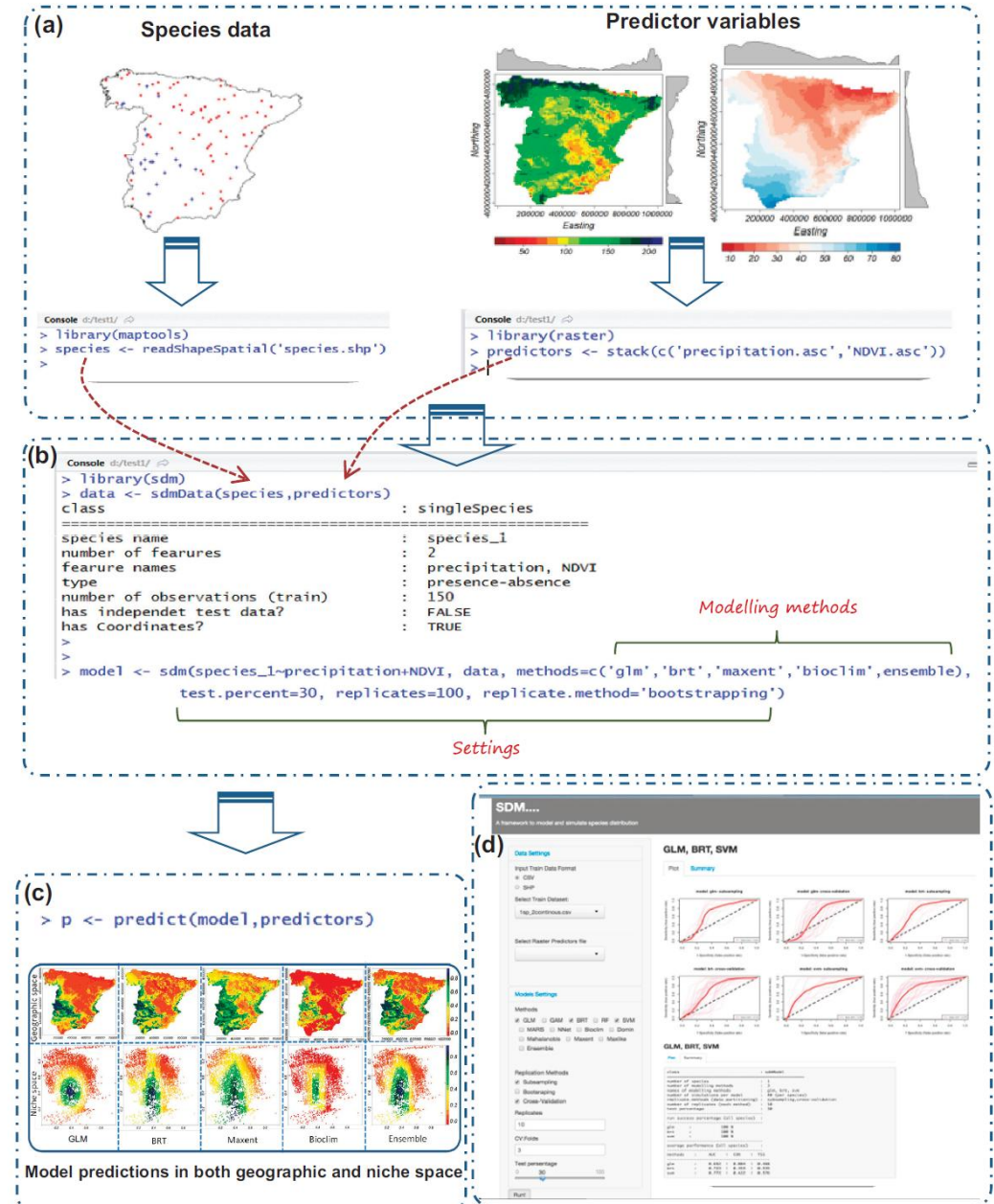


Fig. 6. Examples of response curves for different statistical approaches used to model distribution of plants and vegetation. (a) Generalized linear model with second order polynomial terms; (b) generalized additive model with smoothed spline functions; (c) classification tree; (d) environmental envelope of the BIOCLIM type; (e) canonical correspondence analysis; (f) Bayesian modeling according to Aspinall (1992); p_p = posterior probability of presence of the modeled species, p_{pp} = a priori probability of presence, p_{pa} = a priori probability of absence, p_{cp} = product of *conditional* probability of presence of the various predictor classes, p_{ca} = product of *conditional* probability of absence of the various predictor classes.

Ensemble modelling: One target – multiple models

- Modelling the same target (species) with different methods
- Comparing and combining predictions
- Simultaneous/parallel validation and evaluation
- Powerful tools in R



Joint species distribution modelling

One model – multiple targets

ECOLOGY LETTERS

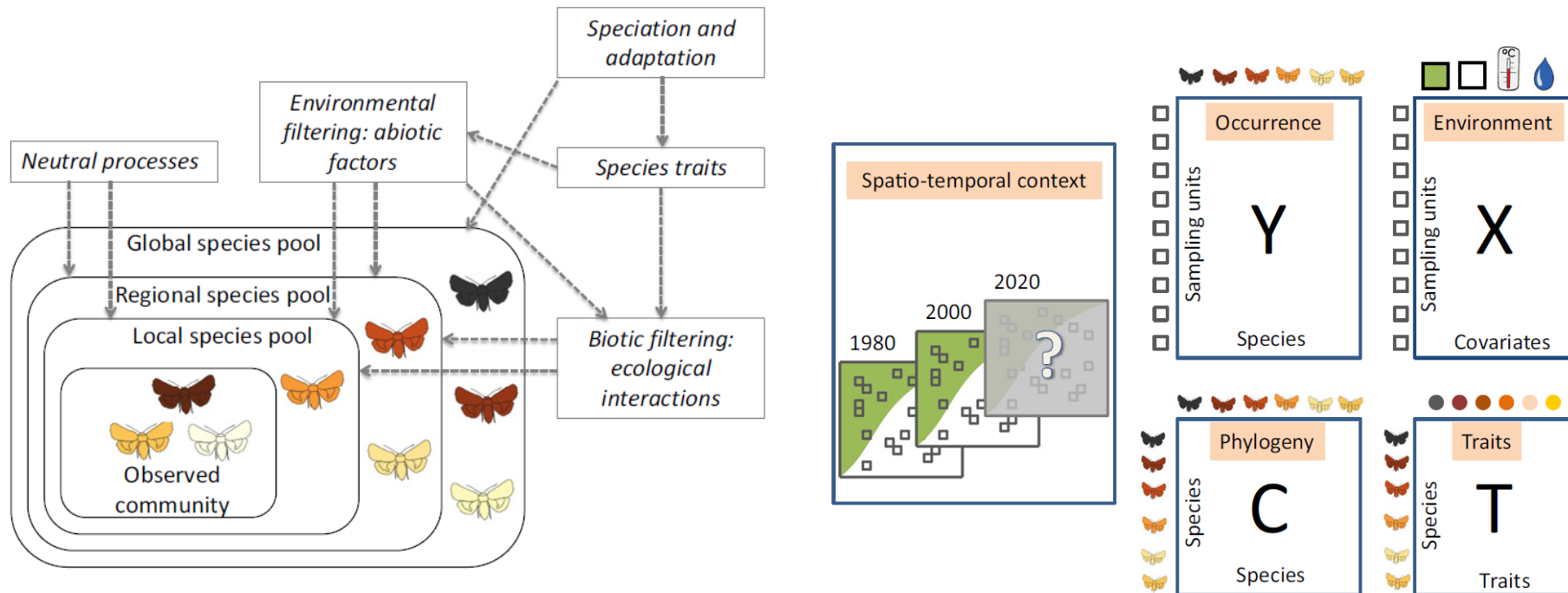
Ecology Letters, (2017) 20: 561–576

doi: 10.1111/ele.12757

IDEA AND PERSPECTIVE

How to make more out of community data? A conceptual framework and its implementation as models and software

Ovaskainen et al. 2017



Mechanistic distribution model example: Integral projection model (IPM)

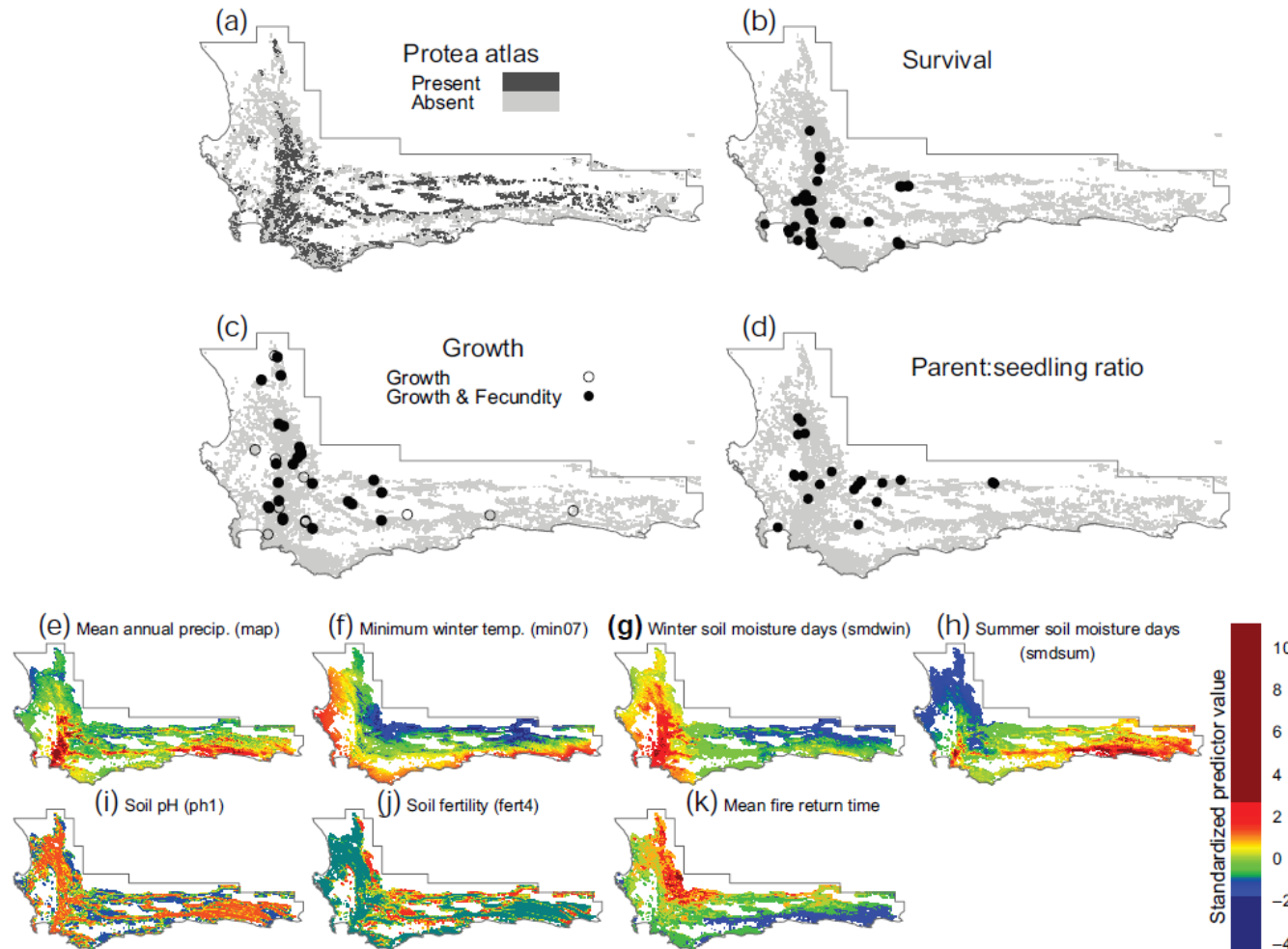
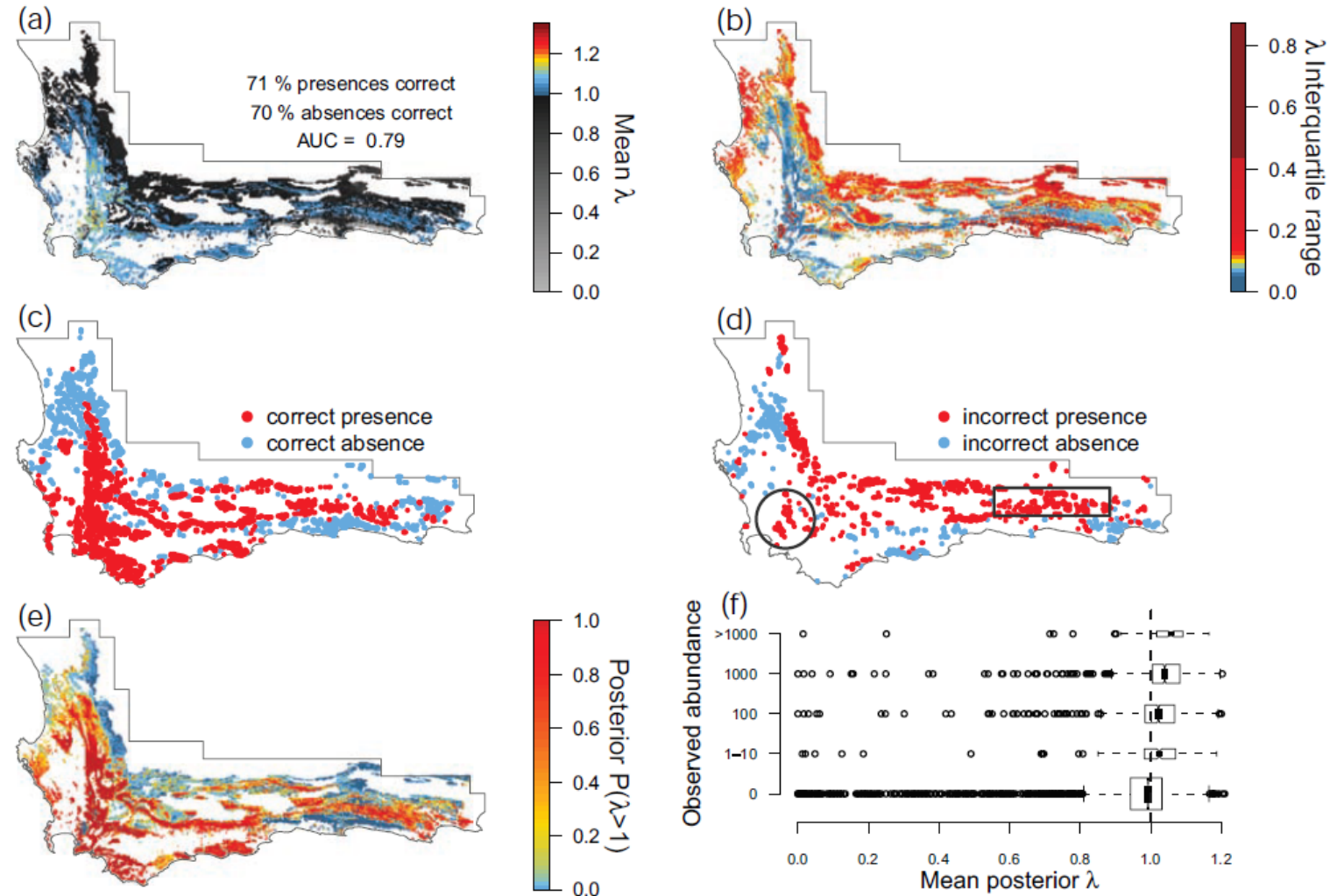


Table 1. Regression coefficients for all vital rate models. Environmental predictors have been standardized. Columns correspond to posterior mean parameter values and lower (upper) 95 percent credible interval bounds. Models were chosen by backward stepwise DIC selection described in Supplementary material Appendix A.

	Mean	Lower-95% CI	Upper-95% CI
Seedling survival probability			
Intercept	-2.597	-3.680	-1.376
Size	4.744	2.807	6.357
Minimum July temperature	-0.362	-0.603	-0.101
Mean annual precipitation	0.188	0.014	0.363
Adult survival probability			
Intercept	3.406	2.782	4.010
Size	-0.695	-0.889	-0.491
Minimum July temperature	0.154	-0.034	0.327
Mean annual precipitation	0.608	0.414	0.846
Growth			
Intercept	0.182	0.167	0.198
% High fertility soil	-0.030	-0.051	-0.013
% High fertility soil ²	0.056	0.018	0.085
Winter soil moisture days	0.036	0.015	0.053
Winter soil moisture days ²	-0.015	-0.024	-0.005
% Acidic soil	-0.002	-0.009	0.007
% Acidic soil ²	-0.035	-0.051	-0.019
Minimum July temperature	0.018	0.011	0.026
Summer soil moisture days	-0.003	-0.012	0.006
Summer soil moisture days ²	-0.030	-0.037	-0.022
Flowering probability			
Intercept	-2.629	-3.755	-1.503
Size	3.627	2.583	4.584
Seedheads/individual			
Intercept	1.495	1.211	1.824
Size	0.622	0.466	0.788
Minimum July temperature	-0.272	-0.401	-0.156
% Acidic soil	0.347	0.163	0.540
% Acidic soil ²	-0.428	-0.652	-0.236
Seeds/seedhead			
Intercept	4.156	4.075	4.241
Germination probability			
Intercept	0.010	0.009	0.012
Offspring size			
Intercept	0.163	0.137	0.193
% Acidic soil	-0.020	-0.035	-0.002
% High fertility soil	-0.054	-0.091	-0.019
% High fertility soil ²	0.065	0.012	0.117
Minimum July temperature	0.041	0.024	0.056
Summer soil moisture days	-0.001	-0.012	0.011
Summer soil moisture days ²	-0.035	-0.051	-0.020

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- Richer output than standard IPMs:
 - Survival
 - Growth
 - Reproduction
 - Population growth (and hence expected presence/absence)
 - Sensitivity/elasticity to environmental conditions



Concluding remarks

- There are many SDM methods; two commonly used are:
 - logistic regression (glm) – presence/absence
 - maxent – presence/background
- When selecting a modelling approach, think about aims, assumptions, data and insight (avoid black boxes)
- Consider advanced SDMs (or entirely different approaches) for species/targets with
 - either: challenges like large observation uncertainty, unobserved states/variables, many missing values, strong biotic interactions, etc
 - or: better data than presence or presence-absence (e.g. abundance, demographics, community data)

Time for exercises!

